

GenAliFA Generative AI in Financial Analytics

S. R. Mahadeva Prasanna

Professor, Dept. of DS & AI, IIIT Dharwad
Director, IIIT Dharwad

prasanna@iiitdwd.ac.in

January 19, 2026



Some Good Resources for LRS

- Fridson, M. S., and Alvarez, F. (2022). Financial statement analysis: a practitioner's guide. John Wiley and Sons.
- Statistics and Data Analysis for Financial Engineering by David Ruppert
- Machine Learning for Asset Managers by Marcos López de Prado
- Tsay, R. S. (2005). Analysis of financial time series. John Wiley and Sons.
- De Prado, M. L. (2018). Advances in financial machine learning. John Wiley and Sons.
- Chishti, S. and Barberis, J. (2016). The FINTECH Book: The Financial Technology Handbook for Investors, Entrepreneurs and Visionaries. John Wiley and Sons.



Syllabus of IFA

- Introduction (1 h)
- Data science for finance (2 h)
- Time-series analysis in finance (2 h)
- Machine learning in finance (2 h)
- Deep learning in finance (2 h)
- Generative AI in finance (2 h)
- Principles and ethics in Finance (1 h)

Lecture Outline

- Motivation and Background
- What is Generative AI?
- Applications in Financial Analytics
- Techniques and Models
- Advantages, Challenges, and Future

What is Generative AI?

Definition

Generative AI refers to models that learn data distributions and **generate new data, scenarios, text, or decisions** rather than only predictions.

- Learns underlying patterns
- Generates plausible futures
- Supports reasoning and explanation

GenAI Techniques Used

- Large Language Models (LLMs)
- Generative pre-trained transformers (GPTs)
- Generative Adversarial Networks (GANs)
- Variational Autoencoders (VAEs)
- Reinforcement Learning
- Hybrid AI systems

Why Generative AI in Financial Analytics?

Modern financial systems are:

- Highly volatile and uncertain
- Influenced by rare but extreme events
- Driven by massive unstructured information
- Subject to regulatory and explainability demands

Need: A paradigm that supports uncertainty-aware decision making.

Background: Evolution of Financial Analytics

Financial analytics has evolved through multiple stages:

- **Descriptive Analytics** Historical reporting and financial ratios
- **Predictive Analytics** Time-series models, regression, ML
- **Prescriptive Analytics** Optimization and rule-based decision systems
- **Generative Intelligence (Current)** Scenario generation, explanation, and simulation



Limitations of Traditional Financial Analytics

- Assumes stable statistical patterns
- Relies heavily on historical data
- Produces single-point forecasts
- Struggles with regime changes and crises
- Limited ability to handle unstructured data

Key Limitation

Traditional models extrapolate the past and predict only one future.

Need for Generative AI

Generative AI is motivated by the need to move beyond single-point predictions toward explainable, scenario-driven financial decision support.

Motivation 1: Scenario-Based Decision Making

- Financial decisions are made under uncertainty
- Decision-makers ask:
 - What if interest rates rise sharply?
 - What if markets crash?
 - What if policy priorities change?

Role of Generative AI

Generates multiple plausible future scenarios instead of a single forecast.

Motivation 2: Modeling Rare and Extreme Events

- Financial crises are low-frequency, high-impact events
- Historical data is insufficient to model extremes
- Traditional risk models underestimate tail risk

Generative AI Advantage

Synthetic generation of stress scenarios and tail-risk exploration.

Why Generative AI Now?

- Availability of large financial and textual datasets
- Advances in deep learning architectures
- Scalable cloud and GPU computing
- Emergence of foundation models with reasoning ability

Why Generative AI in Finance?

- Financial systems are complex, uncertain, and data-rich
- Traditional analytics focuses on prediction
- Modern finance requires:
 - Scenario exploration
 - Risk-aware decision-making
 - Explainability

Key Shift:

Predictive Analytics → Generative Intelligence

Generative AI in Financial Analytics

- Scenario generation instead of point forecasts
- Synthetic data for stress testing
- Natural language reasoning over financial data
- Decision support for investors, firms, and governments

Core Question:

"What could happen, why, and what should we do?"

Key Applications

- 1 Financial Forecasting & Scenario Simulation
- 2 Algorithmic Trading & Market Intelligence
- 3 Risk Management & Stress Testing
- 4 Financial Text Analytics
- 5 Personalized Financial Advisory

Forecasting and Scenario Generation

- Generates multiple plausible future paths
- Captures tail risks and rare events
- Used for:
 - Portfolio stress testing
 - Economic policy simulation

Example:

Simulating market behavior under recession or interest-rate shocks

- Strategy generation and optimization
- Market sentiment synthesis from:
 - News
 - Earnings calls
 - Social media
- Reinforcement learning combined with GenAI

- Credit risk scenario generation
- Fraud pattern synthesis
- Modeling extreme events beyond historical data

Benefit

Improved preparedness for black-swan events

Financial Text Analytics

- Automated analysis of:
 - Annual reports
 - Regulatory filings
 - Earnings call transcripts
- Converts text into:
 - Financial insights
 - Risk indicators
 - Strategic summaries

Traditional vs Generative Analytics

Aspect	Traditional	Generative
Output	Single prediction	Multiple scenarios
Data type	Structured	Structured + unstructured
Explainability	Limited	Natural language
Adaptability	Static	Adaptive

Challenges and Risks

- Hallucination and unreliable outputs
- Regulatory compliance
- Model risk and bias
- Data privacy and ethics

Important

Human-in-the-loop is essential in finance

Future Outlook

- Finance moving from:
 - Predictive → Prescriptive → Proactive
- AI as a co-analyst, not a replacement
- Strong demand for AI + finance expertise

Key Takeaway

Summary

Generative AI transforms financial analytics from forecasting numbers to generating intelligence for informed decision-making.

Thank You

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